

اصول طراحی پایگاه داده

دانشکده کامپیوتر

مدرس: آرش همتی

Database Systems

References

Database System Concepts
Abraham Silberschatz, Henry F. Korth, S. Sudarshan

Fundamentals of Database Systems
Ramez Elmasri, Shamkant B. Navathe

An Introduction to Database Systems
C J Date

Section 0

Data
Information
Knowledge
Wisdom

Data, Information, Knowledge, Wisdom

- Forming the foundations for understanding
- how experience is formed from evidence and observations.
- When data are put into context, the result is information and further contribute to increased knowledge.
- perceiving the source of data enables us to recognize inherent error and bias.

Data are discrete, unprocessed raw set of values (datum), for example quantitative or qualitative facts or observations. Data merely exists; it is immutable and by itself lacks meaning or significance. For example, ‘98, 75, 60, 72, 168’ are data. Although data and information are frequently used interchangeably, the difference is distinct.

Data, Information, Knowledge, Wisdom

Information is data that has been given additional meaning by providing context and is therefore defined by data. For example, the previous slide data can be transformed by describing it as a list of weights from adult male patients attending an orthopaedic day surgery list: ‘98kg, 75kg, 60kg, 72kg, 168kg’.

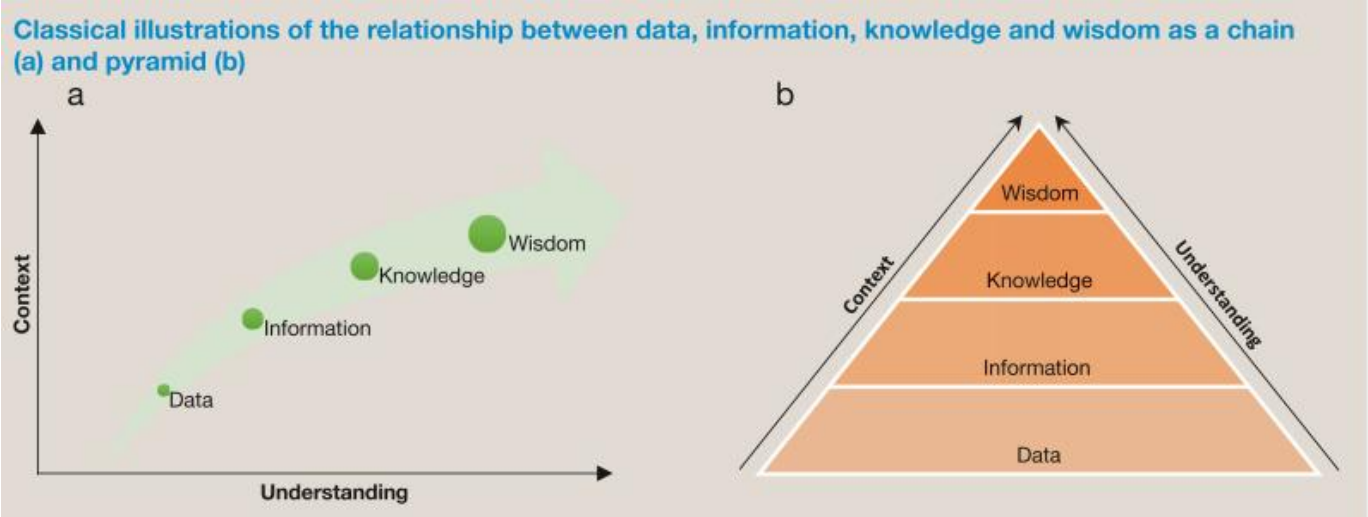
Knowledge is surprisingly difficult to formally define. It does not exist in isolation (i.e. knowledge itself cannot be discovered) but instead is defined in terms of information and allows us to further transform this information into instruction. This is an iterative process whereby study and experience can shape and then reshape knowledge based on observed evidence. Knowledge can be thought of as algorithmic and can be implicit or explicit. Implicit or tacit knowledge is personal and thus difficult to transmit between individuals. It is learned and represents our own experiences. Explicit knowledge is that gained from study and practice and, within a healthcare setting, is contained within protocols and guidelines. Explicit knowledge can be freely passed on. From our example day surgery list, we can reason with knowledge that a 15 kg patient is unlikely to be an adult male.

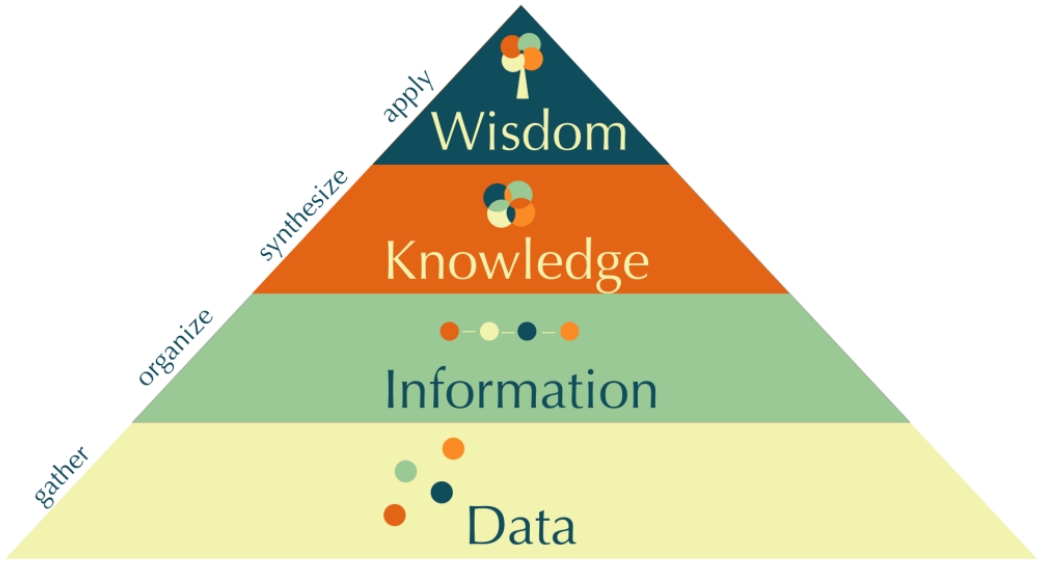
Data, Information, Knowledge, Wisdom

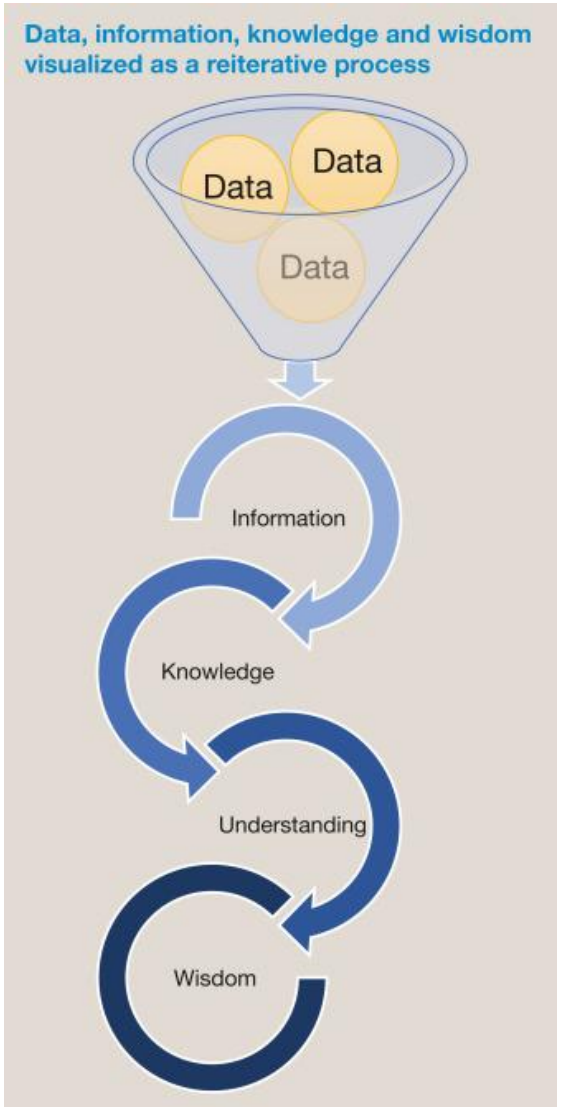
Wisdom is a construct that adds value to knowledge and incorporates individual (and societal) morals and ethics; it is rational, extrapolative, heuristic and non-deterministic. Wisdom endures and is defined in terms of knowledge. For example, it is our wisdom that reasons the 168 kg patient from our example list may not be a suitable day surgery candidate. We sometimes lose sight of wisdom in the search for knowledge and need to be conscious of this. Wisdom concerns the future and enables us to extrapolate, whereas data and knowledge usually refer to the past.

Understanding is considered the overall answer to why and has classically been considered the endpoint of the DIKW model and is arrived at by gaining context and transforming data through to wisdom. In a modified model, understanding provides constant input to wisdom, and vice versa, in a gestalt process; this explains how our knowledge and wisdom change over time based on our experiences (not dissimilar to Bayes' theorem). This facilitates our problem-solving ability: "the key to good decision making is not knowledge… it is understanding". For example, our 168 kg patient is unlikely to be suitable for day surgery as he is likely comorbid, may have obstructive sleep apnoea, anaesthesia and the conduct of surgery itself may not be straightforward and he may need a prolonged recovery and admission.

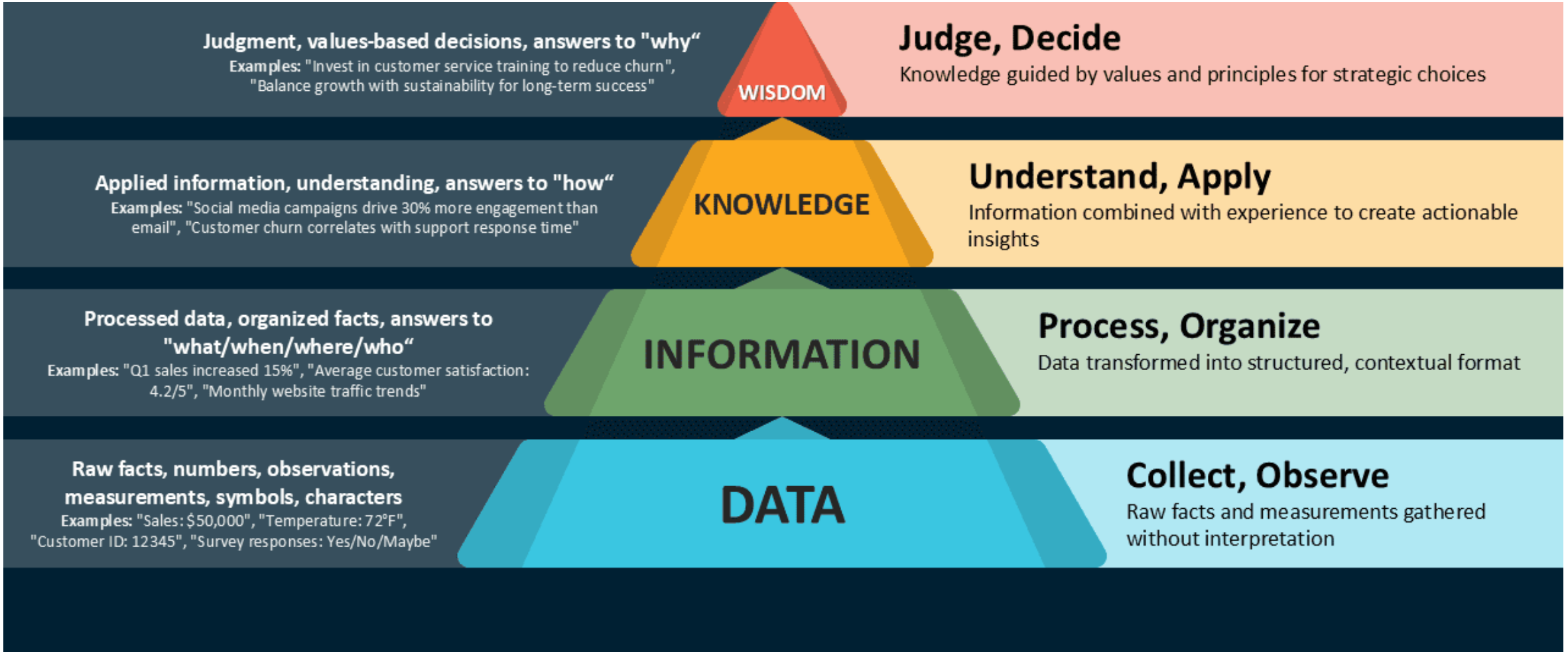
DKIW Pyramid and Flow







DKIW Pyramid



Section 1

Introduction

Intro

Database: The collection of data, usually referred to as the database
DBMS: A database-management system (DBMS) is a collection of interrelated data and a set of programs to access those data.
Database systems are designed to manage large bodies of information. The earliest database systems arose in the 1960s.

It might be claimed that almost all of the softwares in the world have worthy data and might use a database system.

- **Sales**
- **Accounting**
- **Human Resource (HR)**
- **Manufacturing**
- **Banking**
- **Finance**
- **Credit Card Transactions**
- **Universities**
- **Airlines**
- **Telecommunication**
- **Social Media**
- **Online Retailer**
- **Document Systems**
- **Navigation Systems**

Information System Types

- OLTP (Online Transactional Processing)
- OLAP (Online Analytical Processing)

Data Storing Models

- File Base Model
- Database Model

major disadvantages of File Base Model:

- Data redundancy and inconsistency
- Difficulty in accessing data
- Data isolation
- Integrity problems
- Atomicity problems
- Concurrent-access anomalies
- Security problems

Data Model

Underlying the structure of a database is the data model: a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.

- **Relational Model (by Edgar F. Codd)**
- **Entity Relationship Model**
- **Semi-Structured Data Model**
- **Object Based Data Model**
- **Network Model**
- **Hierarchy Model**

Relational Model is most attended model in database design.

Most Prominent RDBMSes

Name	Vendor	Fist Stable Release	License	Model
Oracle Database	Oracle Corp.	1979	Proprietary (OTN)	Relational, Converged
SQL Server	Microsoft	1989	Proprietary	Relational
MySQL	Oracle Corp.	1995	GPLv2	Relational
PostgreSQL	Berkeley Univ.	1996	Free and Open Source	Relational
DB2	IBM	1983	Proprietary	Relational, Converged
H2 (HSQL)	Thomas Mueller	2005	Eclipse Public License	Relational
SQLite	Richard Hipp	2000	Public Domain	Relational
Derby(JavaDB)	Cloudspace-IBM-Apache	1996	Apache License	Relational

NoSQL Data Models

- Document Store
- Key-Value Store
- Graph Store
- Wide-Column

Name	Model
Mongo DB	Document Store
Oracle NoSQL Database	Key-Value
Redis	Key-Value
Apache Cassandra	Graph Store
Apache HBase	Graph Store
Google Cloud Bigtable	Graph Store
Oracle Spatial and Graph	Graph Store

Characteristics of the Database approach

1. Self-Describing Nature of a Database System : Data Catalog (Metadata)
2. Insulation between Programs and Data, and Data Abstraction
3. Support of Multiple Views of the Data
4. Sharing of Data and Multiuser Transaction Processing

Program-Data Independence

In traditional file processing, the structure of data files is embedded in the application programs, so any changes to the structure of a file may require changing all programs that access that file. By contrast, DBMS access programs do not require such changes in most cases. The structure of data files is stored in the DBMS catalog separately from the access programs. We call this property program-data independence.

Program-Operation Independence

In some types of database systems users can define operations on data as part of the database definitions. An operation (also called a function or method) is specified in two parts. The interface (or signature) of an operation includes the operation name and the data types of its arguments (or parameters). The implementation (or method) of the operation is specified separately and can be changed without affecting the interface. User application programs can operate on the data by invoking these operations through their names and arguments, regardless of how the operations are implemented. This may be termed program-operation independence.

Actors on the scene

1. Database Administrators
2. Database Designers
3. End User
4. System Analyzers
5. Application Programmers

Advantages of using DBMS Approach

1. Controlling Redundancy
2. Restricting Unauthorized Access
3. Providing Persistent Storage for Program Objects
4. Providing Storage Structures and Search Techniques for Efficient Query Processing
5. Providing Backup and Recovery
6. Permitting Inferencing and Actions
7. Representing Complex Relationships among Data
8. Enforcing Integrity Constraints
9. Using Rules and Triggers

Data Abstraction

1. Physical level (Internal)
2. Logical Level (Conceptual)
3. View Level (External)

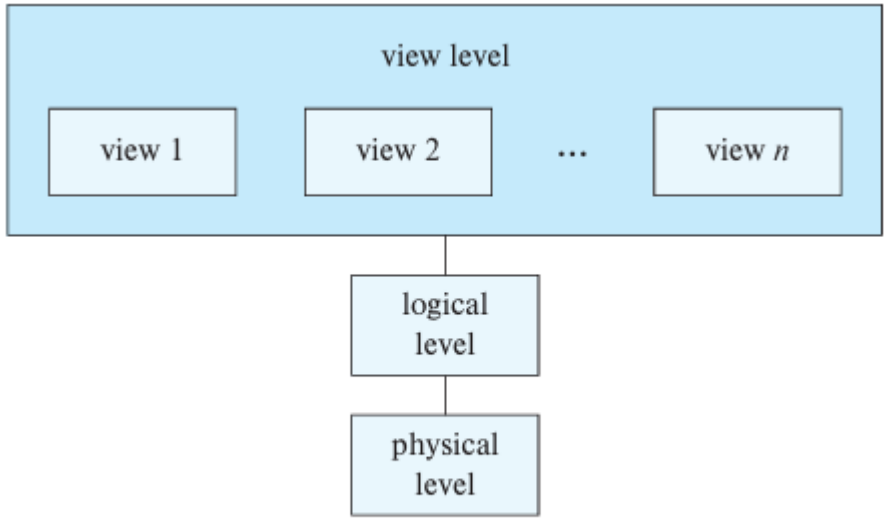
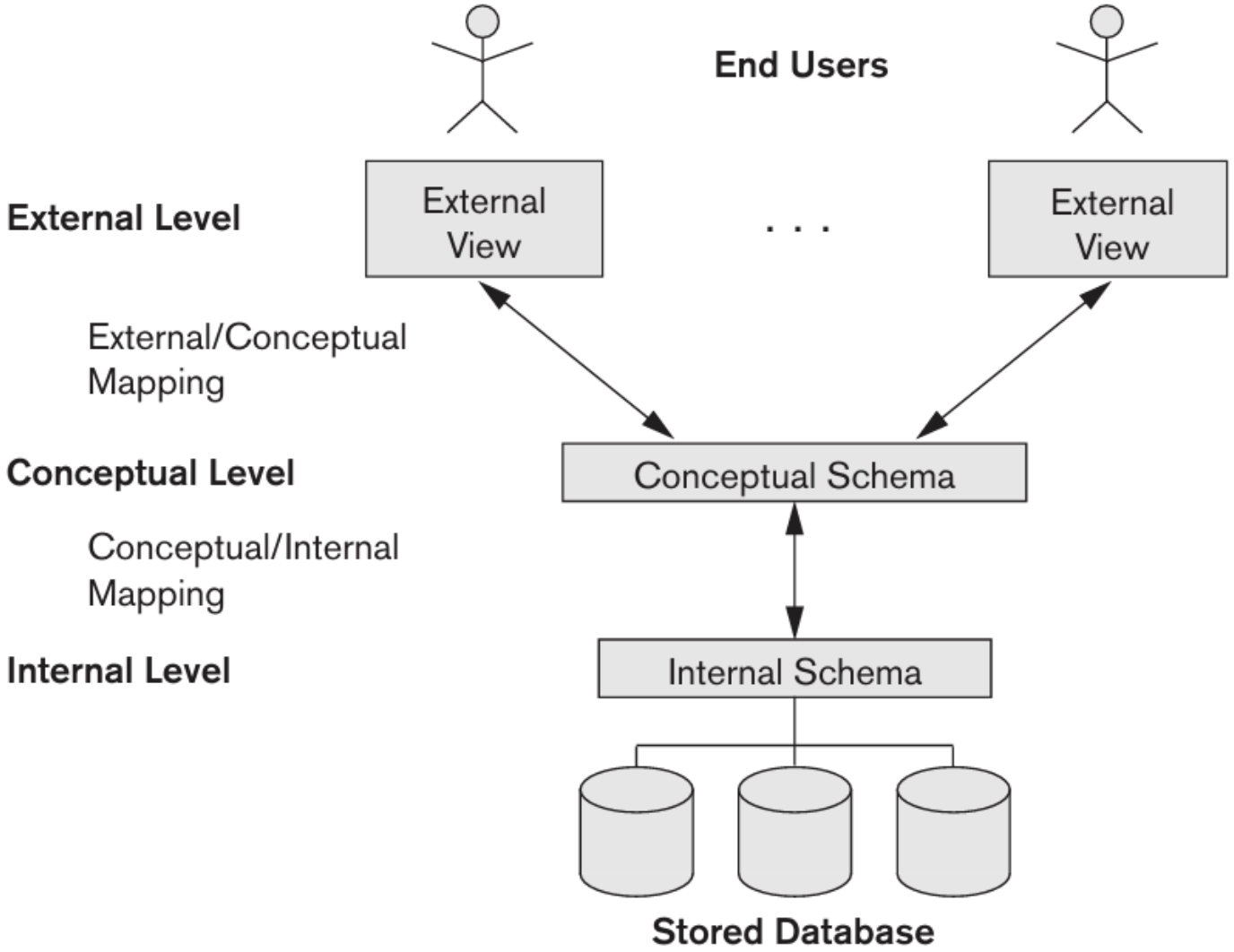


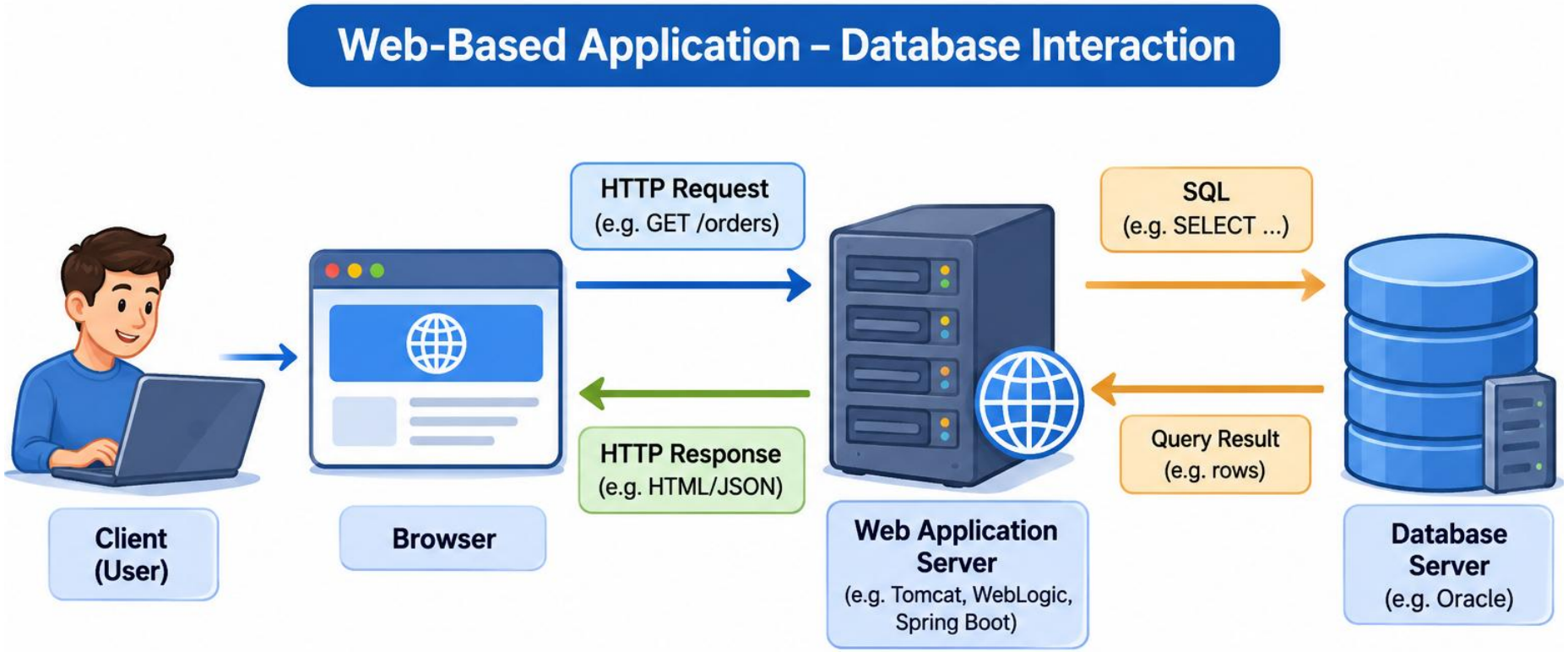
Figure 1.2 The three levels of data abstraction.



Section 2

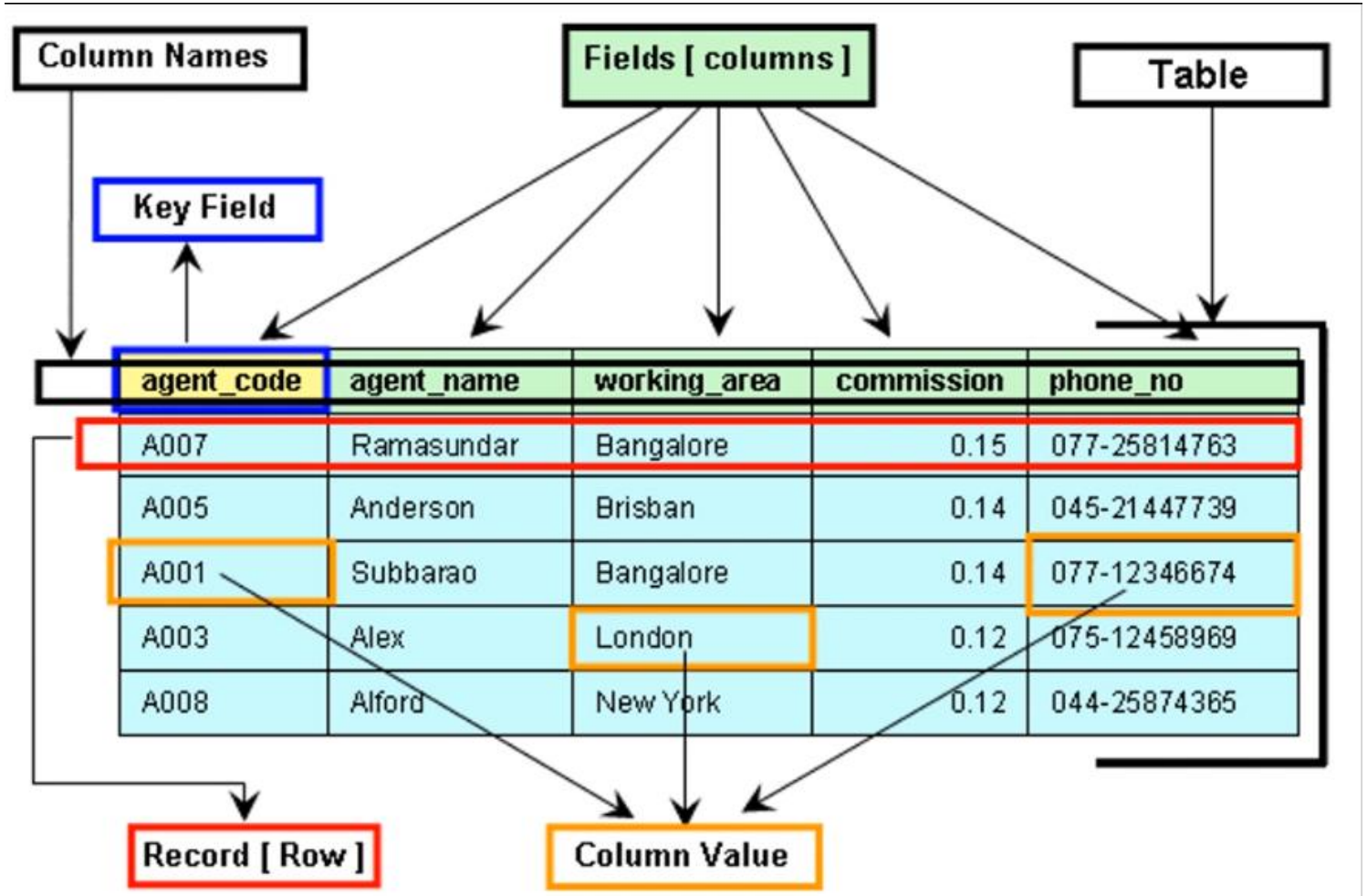
Database System Architecture

Architecture



1. Client uses a browser to access the web application.
2. Browser sends an HTTP request to the web server.
3. Web server processes the request and executes SQL on the database.
4. Database returns the query result to the web server.
5. Web server sends an HTTP response back to the browser.

Table Structure



SQL (Structured Query Language)

A 4GL language used to manage data in a relational database management system.
Introduced in the 1970s, to discard two read-write APIs , ISAM and VSAM.
SQL was initially developed at IBM by Donald D. Chamberlin

In the late 1970s, Relational Software, Inc. (now Oracle Corporation) saw the potential of the concepts described by Codd, Chamberlin, and Boyce, and developed their own SQL-based RDBMS with aspirations of selling it to the U.S. Navy, Central Intelligence Agency, and other U.S. government agencies. In June 1979, Relational Software introduced one of the first commercially available implementations of SQL, Oracle V2 (Version2) for VAX computers.

By 1986, ANSI and ISO standard groups officially adopted the standard "Database Language SQL" language definition. New versions of the standard were published in 1989, 1992, 1996, 1999, 2003, 2006, 2008, 2011, 2016 and, most recently, 2023.

SQL (Structured Query Language)

- DQL (Data query language)
  - SELECT
- DML (Data Manipulation Language)
  - INSERT, UPDATE, DELETE
- DDL (Data Definition Language)
  - CRETAE, DROP
- DCL (Data Control Language)
  - GRANT, REVOKE
- TCL (Transaction Control Language)
  - COMMIT, ROLLBACK

SQL (Structured Query Language)

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- DCL (Data Control Language)
  - GRANT, REVOKE
- TCL (Transaction Control Language)
  - COMMIT, ROLLBACK

Transaction

A transaction is a logical unit of work consisting of one or more database operations (such as reads, writes, updates, or deletes) that must be executed as a single, indivisible unit. Either all operations in the transaction are successfully completed and permanently applied to the database, or none of them are applied at all.

```
BEGIN TRANSACTION;
  UPDATE accounts SET balance = balance - 100 WHERE id = 1;
  UPDATE accounts SET balance = balance + 100 WHERE id = 2;
COMMIT;
```

Transaction: ACID Properties

Property	Description
Atomicity	All operations in a transaction are executed as one indivisible unit — all or nothing.
Consistency	A transaction takes the database from one valid state to another, preserving all defined rules and constraints.
Isolation	Concurrent transactions do not interfere with each other; intermediate states are invisible to other transactions.
Durability	Once a transaction commits, its changes are permanently saved, even in the event of a system crash.

Data Integrity

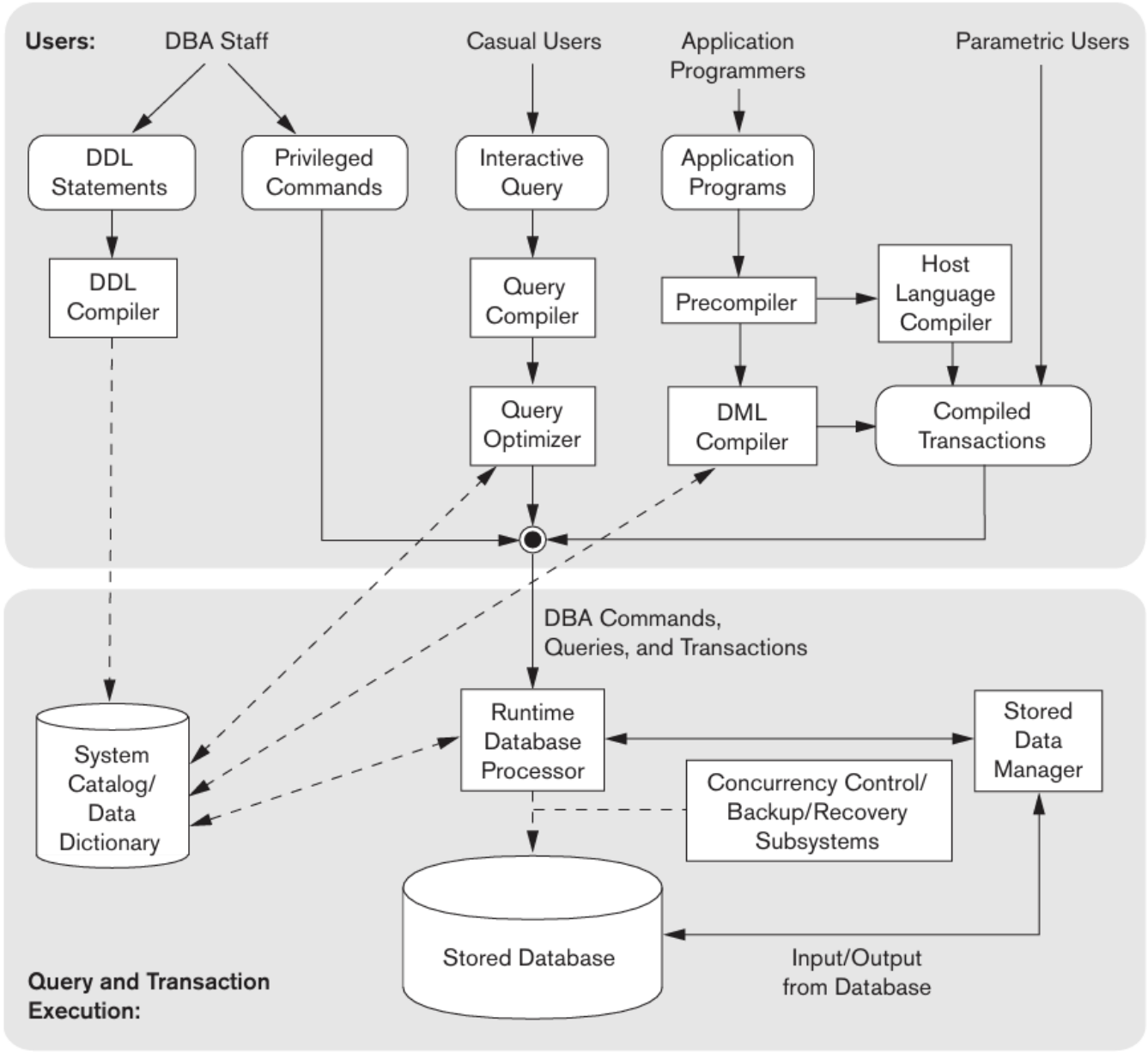
Integrity refers to the accuracy, correctness, and validity of data stored in a database. It ensures that data is reliable and conforms to a set of rules enforced by the database management system (DBMS) throughout its entire lifecycle — from insertion to updates and deletion.

Validity: Data conforms to the defined format, type, range, or domain rules — regardless of whether it reflects reality.

Accuracy: Data conforms to the defined format, type, range, or domain rules — regardless of whether it reflects reality.

Data is free from errors and consistent with all applicable rules, relationships, and real-world facts simultaneously. Correctness is the combination of validity + accuracy + internal consistency. E.g. some data is missed

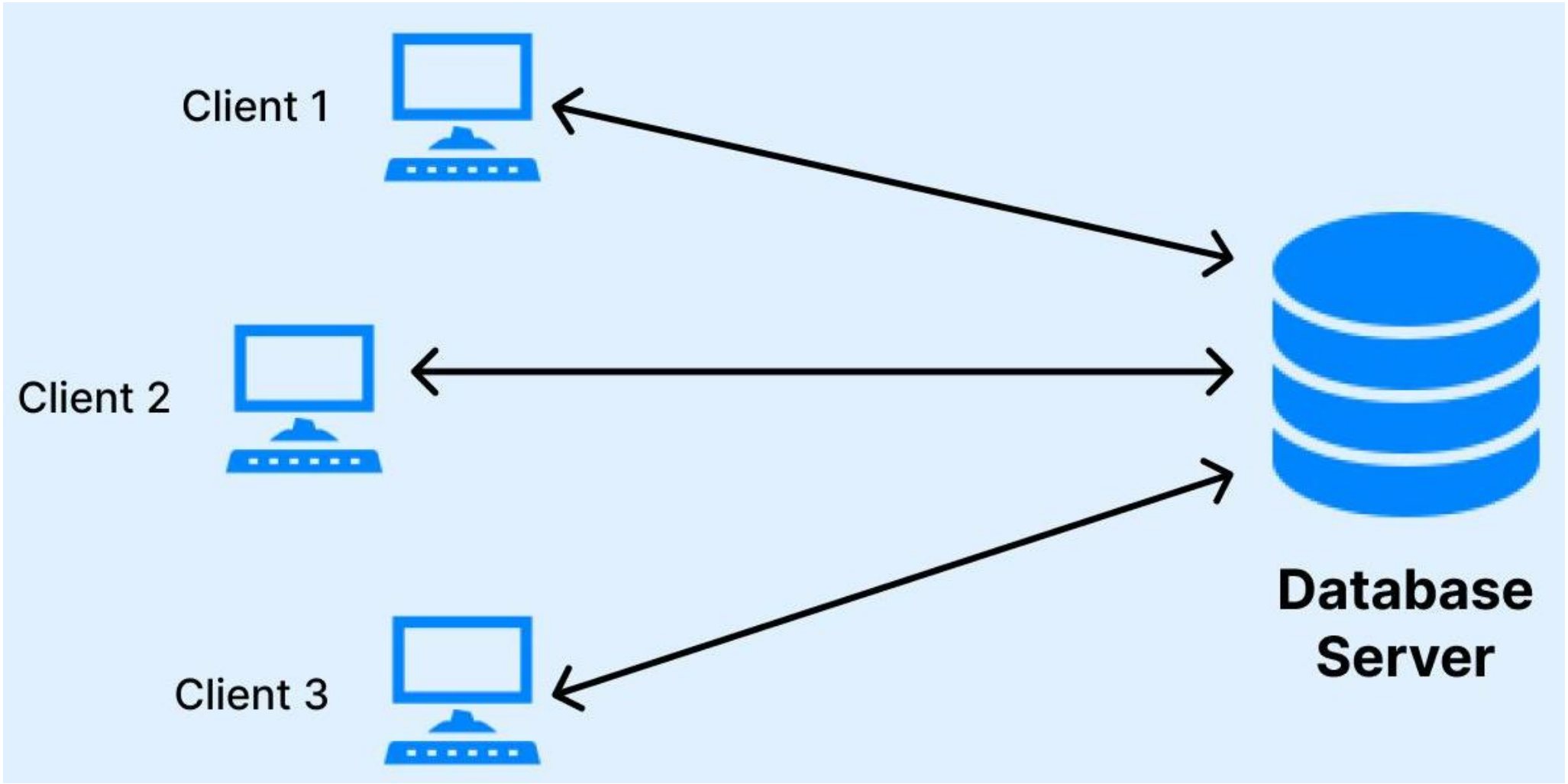
DBMS Components



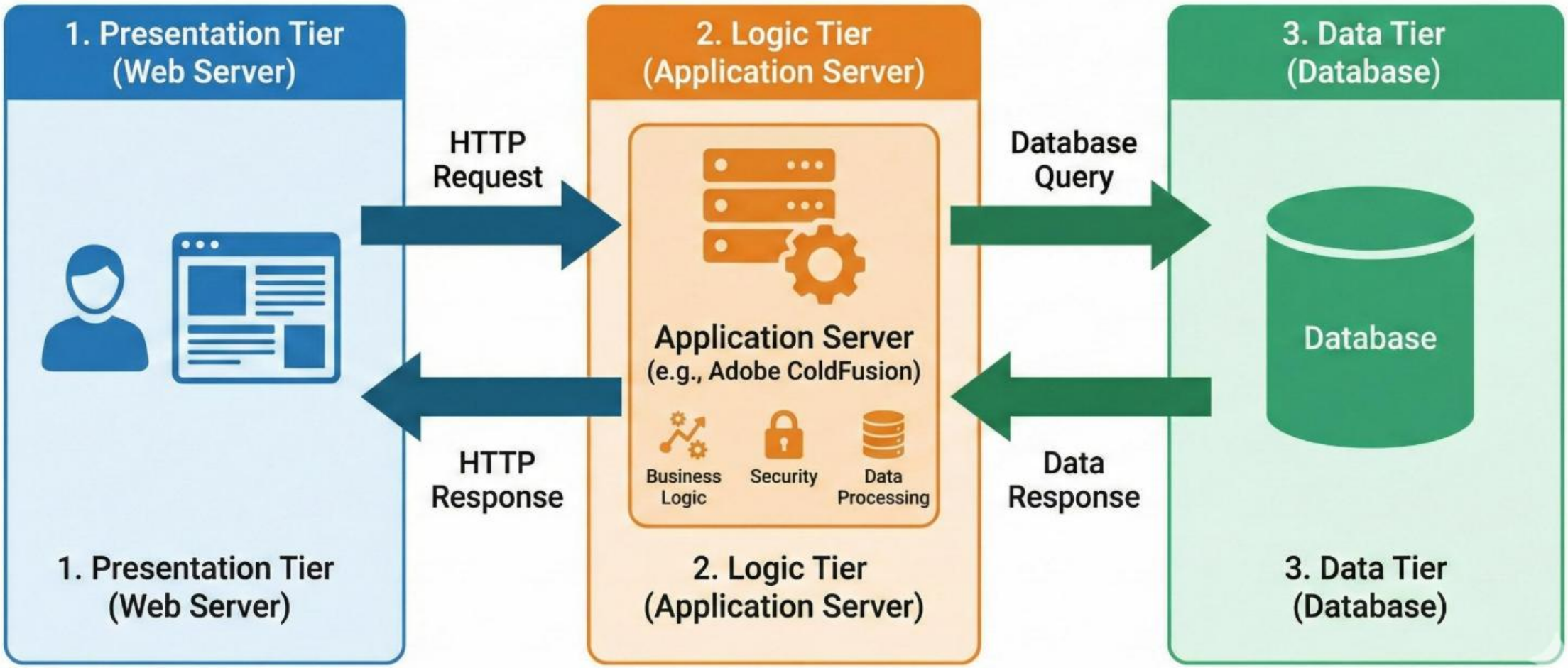
Database System Utilities

- Data Loading
- Backup and Recovery
- Database Storage Reorganization
- Performance Monitoring
- User Auditing
- Import/Export Mechanism
- Job Scheduler

Database Client/Server Architecture



Database Client/Server Architecture



Section 3

Data Modeling

Data Integrity (جامعیت داده ها)

جامعیت داده ها به حفظ صحت، سازگاری و کامل بودن داده ها اشاره دارد. به بیان دیگر حفظ جامعیت داده ها، یعنی جلوگیری از اینکه اطلاعات از بین بروند، یا تغییرات غیر مجاز بر آنها صورت گیرد و اینکه داده ها منطقا اشتباه شوند.

- Accuracy (صحت داده ها)
- Completeness (کامل بودن داده ها)
- Consistency (سازگاری)
- Validity (معتبر بودن)

Data Integrity

- **Accuracy** (صحت داده ها): داده ها عاری از هرگونه خطایی باشند. مثلا حقوق یک کارمند باید با واقعیت تطابق داشته باشد
- **Completeness** (کامل بودن داده ها) :همه اطلاعات مود نیاز وجود داشته باشد، مثلا اطلاعات سفارش مشتری که آدرس تحویل ندارد کامل نیست
- **Consistency** (سازگاری) :داده ها در کل سیستم منطق یکسانی را دارا باشند. مثلا اگر شماره سفارش یک مشتری تغییر کرد در کل سیستم این تغییر اعمال شود.
- **Validity** (معتبر بودن): داده ها با فرمت های مجاز مطابقت داشته باشند. مثلا نوع بیمه اتومبیل می تواند شخص ثالث یا بیمه بدنه باشد. ولی بیمه بیکاری مقداری غیر مجاز است.

Data Integrity (جامعیت داده ها)

جامعیت داده ها در پایگاه داده توسط مکانیزم های زیر می تواند تا حدی تضمین شود.

- Entity Integrity: using PK in tables
- Domain Integrity: using data types
- Referential Integrity: using FK in tables
- User Defined Integrity: using Check Constraints

Normal Design

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)

1NF

- Primary Key is needed
- Multi value is forbidden (Repeating Group)
- Mixing data types is forbidden
- Implicit Row order is forbidden

1NF

Customer_ID	Customer_Date_of_Birth
1001	14-Apr-1998
1001	21-Oct-2002

Customer_ID	Customer_Favorites
1001	Ice Coffee, Moca
1002	Latte, Espresso

Beatle Name	Height
George	178
Paul	179
Ringo	Near 170 cm
John	180

1NF

Beatles_Height_Ranking
George
Paul
Ringo
John

Beatles_Height_Ranking	Order
George	1
Paul	2
Ringo	3
John	4

Anomaly

- **Insert Anomaly : Player CANE has no item, so is not inserted**
- **Update Anomaly: Pati got a Ring so level up to Advanced**
- **Delete Anomaly: Rino loses his/her Arrow, so it removed from table**

Player_ID(PK)	Item_Type(PK)	Item_Quantity	Player_Rating
Pati	Amulet	2	Intermediate
Pati	Ring	4	Intermediate
Rino	Arrow	1	Beginner
Scorp	Shield	3	Advanced
Scorp	Arrow	10	Advanced
Scorp	Coin	5	Advanced
Scorp	Ring	8	Advanced

Functional Dependency (وابستگی تابعی)

It describes a situation where the value of one attribute (or set of attributes) uniquely determines the value of another attribute (or set of attributes).

In the following relation, we says X determines Y, or Y is dependent to X

$$X \rightarrow Y$$

2NF

- **Should be 1NF**
- **Each non key attribute must depends on entire primary key.**

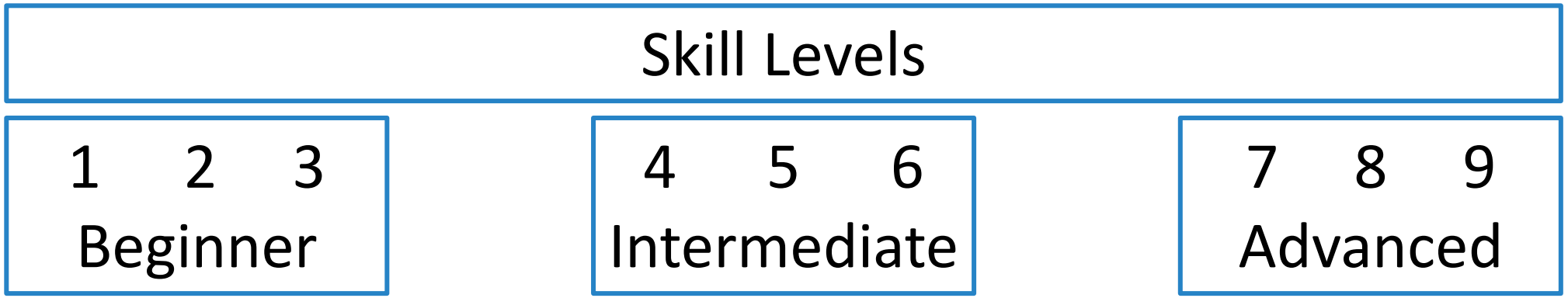
(هر ویژگی غیر کلیدی باید به کل کلید اصلی وابسته باشد نه قسمتی از آن)

$$Player_ID \rightarrow Player_Rating$$

$$Player_ID\ , Item_Type \rightarrow Item\ _Quantity$$

3NF

Player_ID(PK)	Player_Rating	Player_Skill_Level
Pati	Intermediate	4
Tiki	Beginner	4
Rino	Advanced	8
Scorp	Beginner	1



3NF

- **Should be 2NF**
- **Each non key attribute must only and only depends on primary key.**

(هر ویژگی غیر کلیدی باید فقط و فقط به کلید اصلی وابسته باشد و نه چیز دیگری)

$$Player_ID \rightarrow Player_Skill_Level \rightarrow Player_Rating$$

BCNF (More strict version of 3NF)

- **Should be 3NF**
- **Each attribute must only and only depends on primary key.**

(هر ویژگی باید فقط و فقط به کلید اصلی وابسته باشد و نه چیز دیگری)

$$Player_ID \rightarrow Player_Skill_Level \rightarrow Player_Rating$$

از توجه شما سپاسگزارم

پیروز باشید